

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

December 2011

CL103 Mechanics of Solids

Date: 12.12.2011, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Calculate forces in all members of the truss shown in Fig. 1. [07]
- Q - 2 (a) State and prove law of parallelogram. [04]
- (b) Define coplanar forces, collinear forces, concurrent forces & resultant force. [04]
- (c) Find resultant of force system shown in Fig. 2. [06]

OR

- (c) Find the angle between two equal concurrent forces P when their resultant is also equal to P. [06]
- Q - 3 (a) Find centroid of the section shown in Fig. 3. [07]
- (b) A ladder 6m long rests on horizontal ground and leans against a smooth vertical wall at an angle of 20° with the vertical. Its weight is 900 N acting at its middle. It is on the point of sliding when a man weighing 500 N stands on it at a distance 2 m along the ladder from foot of the ladder. Calculate the coefficient of friction. [07]

OR

- Q - 3 (a) Calculate centre of gravity of T-section having flange 200 x 20 mm and web 300 x 20 mm. [07]
- (b) A uniform ladder of weight 'W' and length 'L' is leaning against a vertical wall and horizontal floor. When its inclination is 45° the motion impends. If the wall and the floor are equally rough, find coefficient of static friction, μ_s . [07]

SECTION - II

- Q - 4 (a) Explain how you will carry out tension test on mild steel in the laboratory. Give salient features of tensile test on mild steel. [07]
- Q - 5 (a) Define time period, natural frequency, amplitude and resonance. [04]
- (b) A Mild steel bar of 20mm diameter is acted upon by a tensile force of 60 kN. If the length of bar is 1.2 m, and $E = 2 \times 10^5 \text{ N/mm}^2$, find stress, strain and elongation of the bar. [04]
- (c) A steel bar 1600mm long is acted upon by forces as shown in Fig. 4. Find elongation of the bar if $E = 210 \text{ GN/m}^2$. [06]
- Q - 6 Draw shear force and bending moment diagrams indicating values at all important points for any two beams shown in Fig. 5, Fig. 6 & Fig. 7. Find out point of contra flexure if any. [14]

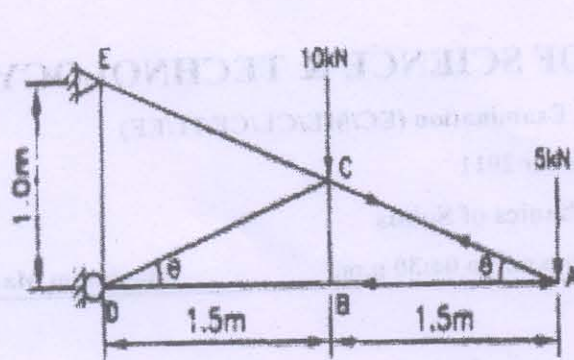


Fig. 1

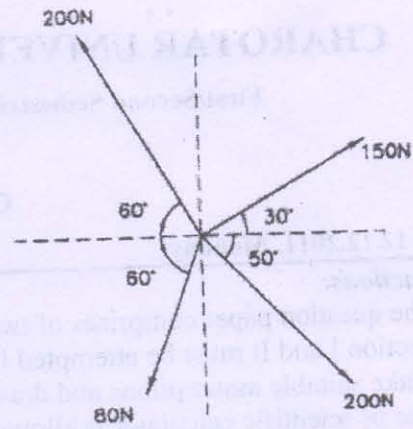


Fig. 2

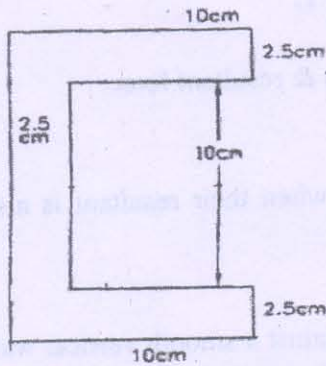


Fig. 3

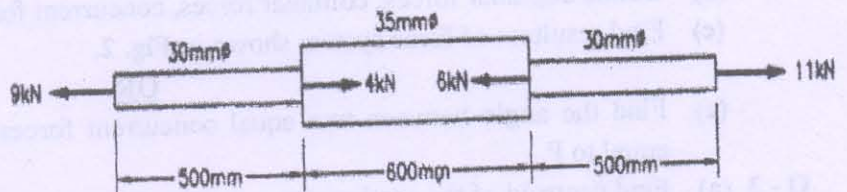


Fig. 4

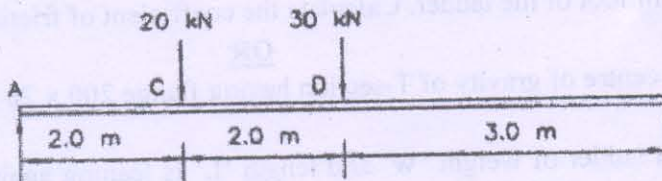


Fig. 5

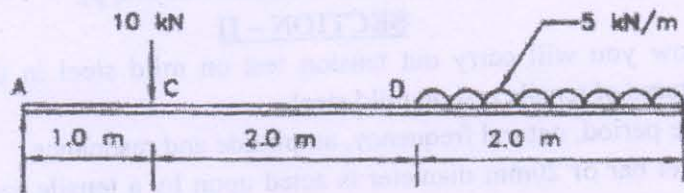


Fig. 6

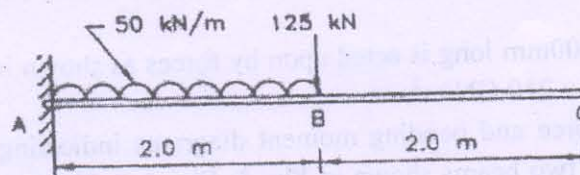


Fig. 7
